



FITBUDDY AI

FITNESS PLAN GENERATOR

Project Documentation • 8-Phase Academic Format

Prepared for SmartBridge Project Submission

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Abstract

FitBuddy AI Fitness Plan Generator is a personalized fitness planning application designed to generate structured workout recommendations based on user-provided information such as fitness goal, experience level, available time, preferred workout style, and available equipment. The system organizes suitable exercises into a practical plan and presents the result in a simple user-friendly interface. The project demonstrates how rule-based logic and AI-oriented personalization can be used to support fitness planning while keeping the application accessible to beginners.

Important Safety Note

FitBuddy AI is intended for general fitness planning and educational use. It is not a medical diagnosis or treatment system. Users with medical conditions or significant exercise limitations should seek advice from a qualified healthcare professional before starting a new exercise program.

Keywords: Artificial Intelligence • Fitness Planner • Personalization • Workout Recommendation • Web Application • User Interface

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IDEATE	ANALYZE	DESIGN	PLAN
→	→	→	→
DEVELOP	TEST	DOCUMENT	DEMONSTRATE

1.1 Problem Statement

Many people want to follow a fitness routine but do not know how to choose exercises, organize workout days, or adapt a plan to their personal goals and available time. Generic workout plans may not match an individual's experience level, preferences, schedule, or equipment. FitBuddy AI addresses this problem by generating a personalized fitness plan from simple user inputs.

1.2 Initial Ideas Considered

- Smart Attendance System – an automated attendance solution using digital identification.
- AI Study Planner – a system that creates personalized study schedules.
- FitBuddy AI Fitness Plan Generator – a personalized workout planning application.
- AI Habit Tracker – an application that recommends routines based on user habits.

1.3 Selected Idea

FitBuddy AI was selected because it combines a practical everyday use case with personalization and artificial-intelligence concepts. The application can be developed as a clear demonstration project with an easy-to-understand input/output workflow.

1.4 Target Users

- Beginners starting a fitness routine.
- Students and young adults looking for structured workouts.
- Users who want simple goal-based workout recommendations.
- People who want a quick plan without manually designing each workout.

2.1 Functional Requirements

- The system shall collect basic fitness preferences from the user.
- The system shall allow selection of a fitness goal such as general fitness, strength, endurance, or flexibility.
- The system shall consider experience level and available workout time.
- The system shall consider equipment availability where applicable.
- The system shall generate a structured workout plan.
- The system shall display exercises, repetitions/time, and workout-day organization.
- The system shall provide a clear reset or regenerate option.

2.2 Non-Functional Requirements

- Usability – the interface should be simple and understandable.
- Performance – plan generation should complete quickly for normal inputs.
- Reliability – invalid or incomplete input should be handled safely.
- Maintainability – exercises and recommendation rules should be easy to update.
- Privacy – unnecessary personal information should not be collected.

2.3 Proposed Technical Stack

Layer	Technology / Option
Frontend	HTML5, CSS3, JavaScript
Backend	Python with Flask (if required)
AI / Recommendation	Rule-based personalization or Python ML logic
Data	JSON / SQLite / CSV, depending on implementation
Development	VS Code / GitHub
Deployment	Localhost or suitable web hosting
Hardware / Software	Computer/laptop with modern browser; internet where required; VS Code; Python environment if a Python backend is used.

3.1 System Architecture

The system follows a simple input–processing–output architecture. The user enters fitness preferences through the interface. The recommendation engine validates the input, applies fitness-plan rules or AI logic, selects suitable exercises, organizes them into a schedule, and displays the generated plan.

USER	WEB / UI	VALIDATION	AI / RULE ENGINE	FITNESS PLAN
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3.2 Main Modules

- User Input Module – collects goal, level, time, equipment, and preferences.
- Validation Module – checks whether required selections are valid.
- Recommendation Engine – maps user requirements to appropriate exercises and schedule.
- Exercise Database – stores exercise names, categories, difficulty, and instructions.
- Plan Generator – creates the final day-wise plan.
- Result Display Module – presents the generated plan clearly.

3.3 Basic Flow

- Start application.
- Enter fitness details.
- Validate inputs.
- Identify goal and experience level.
- Select suitable exercises.
- Arrange exercises into a plan.
- Display the plan.
- Allow the user to regenerate or reset.
- End.

3.4 UI Wireframe Description

HOME	DETAILS FORM	GENERATE PLAN	WEEKLY / DAILY PLAN	RESET
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4.1 Team Roles

Role	Responsibility
Team Lead	Coordinate work, requirements, integration, and presentation.
Developer	Implement frontend/backend and recommendation logic.
UI/UX	Design screens, forms, layout, and usability.
Testing & Documentation	Prepare test cases, screenshots, report, and final documentation.
4.2 Four-Week Sprint Plan	
Week 1	Ideation, requirements, research, technology selection → Problem statement and requirements
Week 2	Architecture, UI design, exercise data, initial coding → Design and prototype
Week 3	Recommendation engine, integration, testing → Working application
Week 4	Bug fixing, documentation, demo recording, GitHub submission → Final project package

5.1 Development Approach

Development begins with the user interface and input form, followed by exercise data organization and recommendation logic. The modules are then integrated and tested together. A modular structure allows exercise data, rules, and interface components to be updated independently.

5.2 Suggested Project Structure

```
FitBuddy-AI/  
  ■■■ frontend/  
    ■■■ index.html  
    ■■■ style.css  
    ■■■ script.js  
  ■■■ backend/  
    ■■■ app.py  
    ■■■ recommender.py  
  ■■■ data/  
    ■■■ exercises.json  
  ■■■ docs/  
    ■■■ ideation.md  
    ■■■ requirements.md  
    ■■■ design.md  
    ■■■ timeline.md  
    ■■■ testing.md  
  ■■■ README.md
```

5.3 Recommendation Logic

- Read user goal, level, available time, and equipment.
- Filter exercises according to suitability and constraints.
- Assign an appropriate number of exercises for the available time.
- Organize exercises across selected workout days.
- Display a clear plan with exercise names and prescribed duration/repetitions.
- Include a general safety note and encourage appropriate professional advice when necessary.

5.4 GitHub Repository

The repository can contain the eight phase folders, source code, README, screenshots, testing evidence, final report, and demonstration link. The repository should be public if the project submission instructions require a public repository.

6.1 Testing Objective

Testing verifies that the application accepts valid inputs, handles invalid or incomplete inputs appropriately, generates a plan, and presents results without interface errors.

6.2 Test Cases

TC ID	Test Scenario	Expected Result
TC01	Open application	Home page loads successfully.
TC02	Submit valid fitness details	Personalized plan is generated.
TC03	Leave required field empty	User receives validation message.
TC04	Change fitness goal	Generated plan reflects the selected goal.
TC05	Change experience level	Exercise difficulty is adjusted appropriately.
TC06	Select limited equipment	Plan avoids unavailable equipment.
TC07	Use regenerate option	A valid plan is generated again.
TC08	Use reset option	Input fields return to initial state.
TC09	Open on mobile-sized screen	Layout remains readable and usable.
TC10	Refresh application	Application loads without breaking.

6.3 Bugs and Fixes

- Incorrect or missing input handling → added validation.
- Exercise mismatch → added goal/level/equipment filtering.
- Layout issues on small screens → added responsive CSS.
- Repeated or unsuitable recommendations → improved selection and filtering logic.

6.4 Testing Evidence

During final submission, add screenshots of the home page, input form, generated plan, validation message, and responsive/mobile view in this section.

7.1 Project Objectives

- Provide an easy-to-use personalized fitness planning interface.
- Demonstrate AI-oriented recommendation and personalization concepts.
- Reduce the effort required to manually organize a workout routine.
- Create a modular project that can be extended with more exercises and smarter recommendations.

7.2 Benefits

- Personalized recommendations.
- Simple and quick plan generation.
- Beginner-friendly interface.
- Can be extended with user progress tracking.
- Can be enhanced with nutrition, reminders, or analytics modules in future versions.

7.3 Limitations

- Recommendations depend on the quality of the exercise data and rules/model.
- The basic version does not replace professional fitness or medical guidance.
- Without wearable/device integration, actual exercise performance is not automatically verified.
- Personalization is limited by the information supplied by the user.

7.4 Future Enhancements

- Progress tracking and workout history.
- Adaptive plans based on completed workouts.
- Optional wearable integration.
- Voice-based exercise guidance.
- Multilingual interface.
- More advanced machine-learning recommendation models.
- Optional nutrition planning with appropriate safety controls.

7.5 Conclusion

FitBuddy AI Fitness Plan Generator demonstrates how software engineering, web development, and AI-based personalization can be combined to create a practical fitness planning application. The project follows a complete development lifecycle from ideation and requirements through design, development, testing, documentation, and demonstration.

8.1 Demonstration Flow (5–6 Minutes)

Time	What to Present
0:00–0:40	Introduce the team, project title, problem statement, and objective.
0:40–1:30	Explain the target users, major features, and technology stack.
1:30–2:30	Open the application and enter sample fitness preferences.
2:30–3:45	Generate the plan and explain how recommendations are organized.
3:45–4:45	Show another input combination and demonstrate personalization.
4:45–5:30	Briefly explain architecture, testing, and GitHub repository.
5:30–6:00	Summarize benefits, future enhancements, and thank the audience.

8.2 Demo Checklist

- Application runs successfully.
- Sample inputs are prepared.
- Generated plan is visible.
- At least one validation case is demonstrated.
- GitHub repository is accessible.
- Demo video is uploaded and the link is added to the repository.
- Final PDF documentation is included.

8.3 Demo Link

Google Drive / YouTube / other approved demonstration link:

FINAL SUBMISSION CHECKLIST

01	■	Public GitHub repository created.
02	■	Eight phase folders/files added.
03	■	Source code uploaded.
04	■	README contains setup and usage instructions.
05	■	Testing screenshots added.
06	■	Final documentation PDF uploaded.
07	■	Demo video link added.
08	■	Project presentation/demo is ready.

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End of Project Documentation